# Bonsai: Edit-Based Preference Learning for Interactive Digital Narrative Authoring

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## Abstract

Authors of LLM-based interactive digital narratives (IDNs) struggle to preserve creative intent as player choices and real-time generation pull storylines in unpredictable directions. We propose *cultivation*: authors edit AI-improvised content rather than specifying intent upfront, and each edit yields natural-language preference rules that steer future generations. We instantiate this in *Bonsai* and evaluate it across three projects using an LLM simulated author. We find that intent rules are project-specific (rules from the wrong author actively hurt performance) and generalize to unseen scenes (outperforming few-shot prompting in 92% of pairwise comparisons). Finally, we show that preference extraction must consider the medium: priming extraction on IDN authoring categories recovers world-constraint preferences almost entirely missed by naive extraction. These results suggest IDN authoring tools can learn author preferences from ordinary edits, providing an inspectable record of intent that grows with use.

## 1. Introduction

Large language models (LLMs) expand the capabilities of interactive digital narratives (IDNs) by granting near-limitless player agency. Systems like *AI Dungeon* cast LLMs as dungeon masters to "play along" with arbitrary player inputs in real-time. But this freedom comes at a cost to authorial influence. LLMs drift, pulled by player tangents and their own improvisation (Jones & Millard, 2024; Mason, 2021), and resist constraint (Toyer et al., 2023). An author who dislikes the new branch finds the LLM unreliable; an author who would have embraced it was never consulted.

Attuning LLMs to authorial intent is a kind of alignment problem. The system must continue narratives coherently across unanticipated inputs (e.g. befriending the dragon rather than slaying it) while retaining author idiosyncrasies across prose, world-building, and interaction mechanics (Wu et al., 2025). Edit-based preference learning (Gao et al., 2024; Liu et al., 2023a; Chakrabarty et al., 2025) offers a natural fit: authors correct generations they dislike, edits accumulate as *intent rules*, and alignment improves over time. Where fine-tuning requires ≈1K curated samples per author (Zhou et al., 2023; Ziegler et al., 2020; Ouyang et al., 2022) and prompting relies on brittle heuristics (Zamfirescu-Pereira et al., 2023), edit-based learning requires only a few examples.

In this paper, we describe *Bonsai*, an LLM-based IDN authoring tool built around *cultivation*, an authoring loop in which these edits emerge as a byproduct of ordinary use. *Cultivation* operates in four phases. An author *seeds* the initial IDN. Players *grow* new branches via interaction; the system generates branches for any inputs unmatched by existing content, adding them to the live artifact. Authors return asynchronously to *prune* and edit the branches that emerged. The system *adapts* to the author's edits, inferring intent rules that accumulate to steer future improvisation. Unlike existing authoring frameworks that treat IDNs as static artifacts once published, *Bonsai* produces an artifact that evolves with author, player, and system.

We evaluate cultivation on three IDN projects using an LLM oracle to simulate author edits. Intent rules outperform few-shot prompting in 92% of pairwise comparisons, generalize to unseen scenes (+2.56 on a 1–5 scale, exceeding the training-scene lift of +1.83), and are project-specific: rules from the wrong author actively hurt (–0.44 vs. no rules). Critically, what edits surface depends on the structure of extraction. Priming with Wu et al.'s IDN authoring categories (world-settings and characters, narrator behavior, and player interactions) improves world-constraint recovery from 8% to 83%. What a creative authoring tool can learn from edits depends on the medium; as IDNs encompass a different scope of preferences than creative writing, the extraction design must be built to match.

We release *Bonsai*—including its custom markup language for IDN authoring and evaluation harness—as open source

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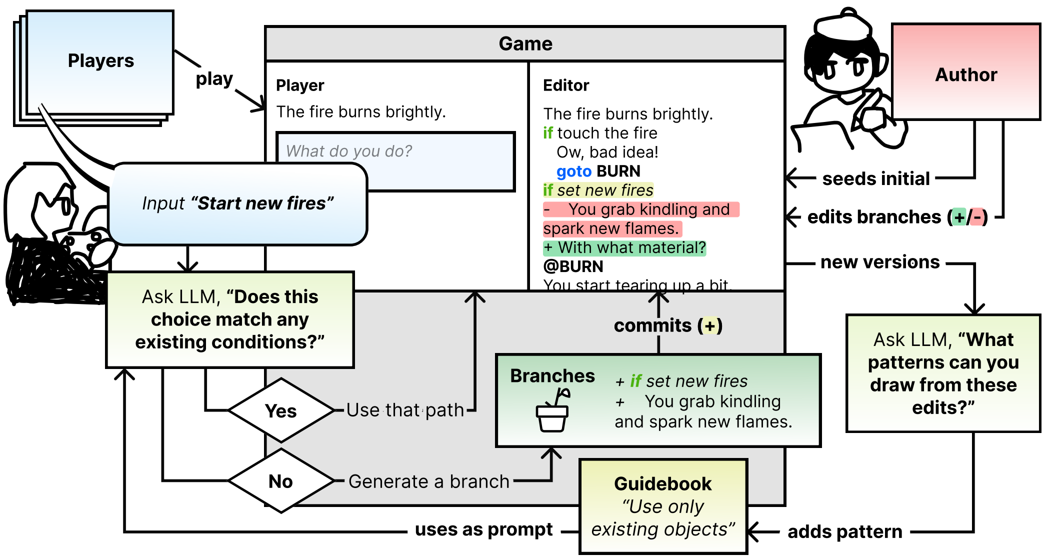


*Figure 1.* Overview of Bonsai's cultivation cycle. Players input natural language actions (e.g., *"Start new fires"*), triggering similarity matching against existing paths. Novel actions generate new branches guided by the accumulated rule set. Authors asynchronously review generated content in the Editor, accepting (+) or rejecting (-) branches. Accepted edits create new versions and add learned patterns to the rule set (e.g., *"Use only existing objects"*), which inform future generation.

to support further work in preference learning for IDN authoring.

## 2. Related Work

Interactive digital narratives (IDNs) grant players what Murray calls "the satisfying power to take meaningful action and see the results of your decisions and choices" (Murray, 1997), and what Mawhorter et al. later formalizes as *choice poetics* (Mawhorter et al., 2014): the meaning constructed through the design of choices themselves, independent of which option the player selects. Both depend on a system the player can trust to behave consistently and an author with enough control to make choices meaningful. Yet player agency is fundamentally limited by what different IDN authoring forms allow.

Conventional branching IDNs (e.g. those implemented via Twine) disallow paths that were not explicitly defined. This means as narrative complexity expands, it becomes combinatorially burdensome to define all paths any player might desire (Jones & Millard, 2024), limiting the degree of comprehensive player agency supported by such a system. While modular content structures like storylets enable easier refactoring and emergent interaction design (Kreminski & Wardrip-Fruin, 2018b; Short, 2016), the author remains solely responsible for defining choices that lead to meaningful story progression.

LLMs offer a potential solution to these challenges through their ability to improvise coherent responses to arbitrary player input. Early systems like AI Dungeon (Ferreira, 2025) use LLMs to generate narrative content on demand. Pure prompt-based generation introduces failure modes: models hallucinate (Kalai et al., 2025), misinterpret inputs (Liu et al., 2023b), get derailed by player tangents (Gallotta et al., 2024), and lack understanding of narrative structure (Chakrabarty et al., 2024; Tian et al., 2024). Authors are not consulted when the LLM improvises a response—even for player-suggested scenarios they would strongly prefer to hand-write or restrict—and the LLM-improvised material is discarded after play, rather than stored, recalled, or refined for later playthroughs. Worse, an author who wants certain narrative paths to be impossible may find the LLM cannot be made to reliably comply (Toyer et al., 2023).

To address these consistency issues, recent systems impose additional structure on LLM generation. *Dramamancer* combines modular storylets with authored preconditions that gate critical state changes while generating narration and dialogue in real-time (Wang et al., 2025). Orchid uses upfront specifications to constrain narrative progression (Wu et al., 2025). These approaches improve coherence over pure prompt-based generation, yet require authors to anticipate player choices and specify all constraints upfront, giving authors no mechanism to learn from or correct what actually emerges across playthroughs.

Beyond IDN, creativity support tools have explored author-system alignment through preference learning from edits (Gao et al., 2024), revision-focused interfaces (Zhou & Sterman, 2024), and multimodal intent capture (Chung et al., 2022; 2025). These improve creative preference alignment in solo creative writing, but none addresses the triadic challenge of IDN authoring: balancing author intent, player agency, and system behavior simultaneously, across multiple playthroughs. No prior work has applied Kreminski & Wardrip-Fruin's gardening orientation (Kreminski & Wardrip-Fruin, 2018a) to IDN authoring with generative AI.

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